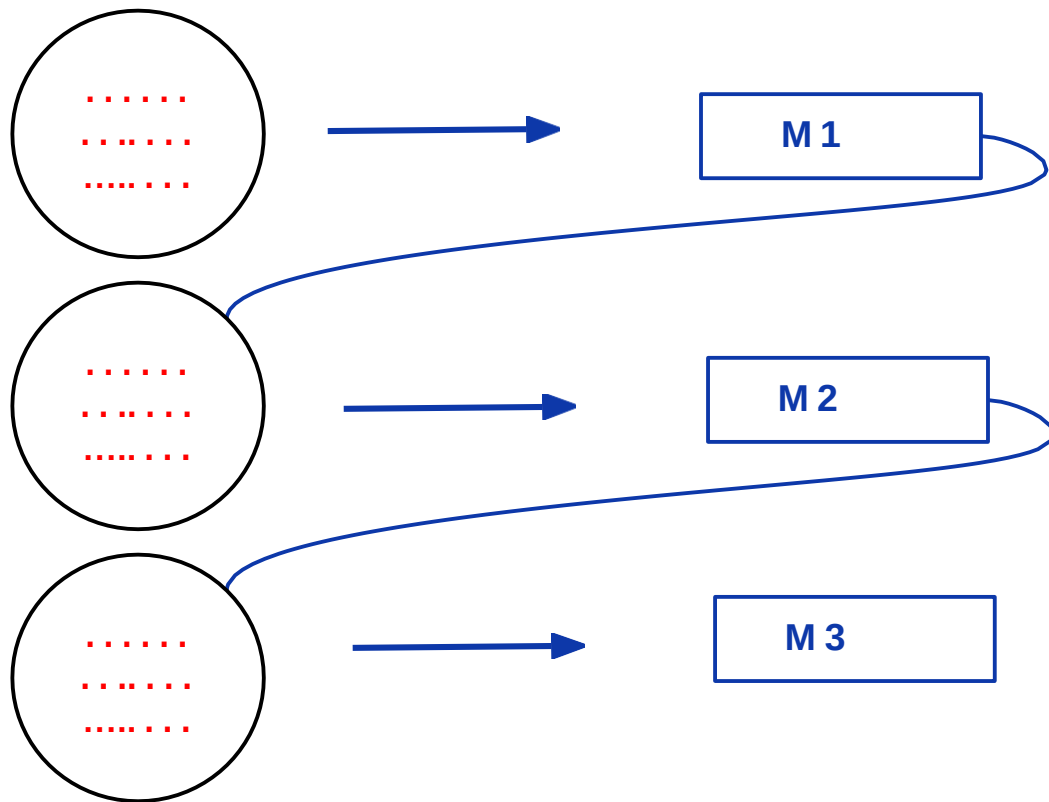


# Boosting



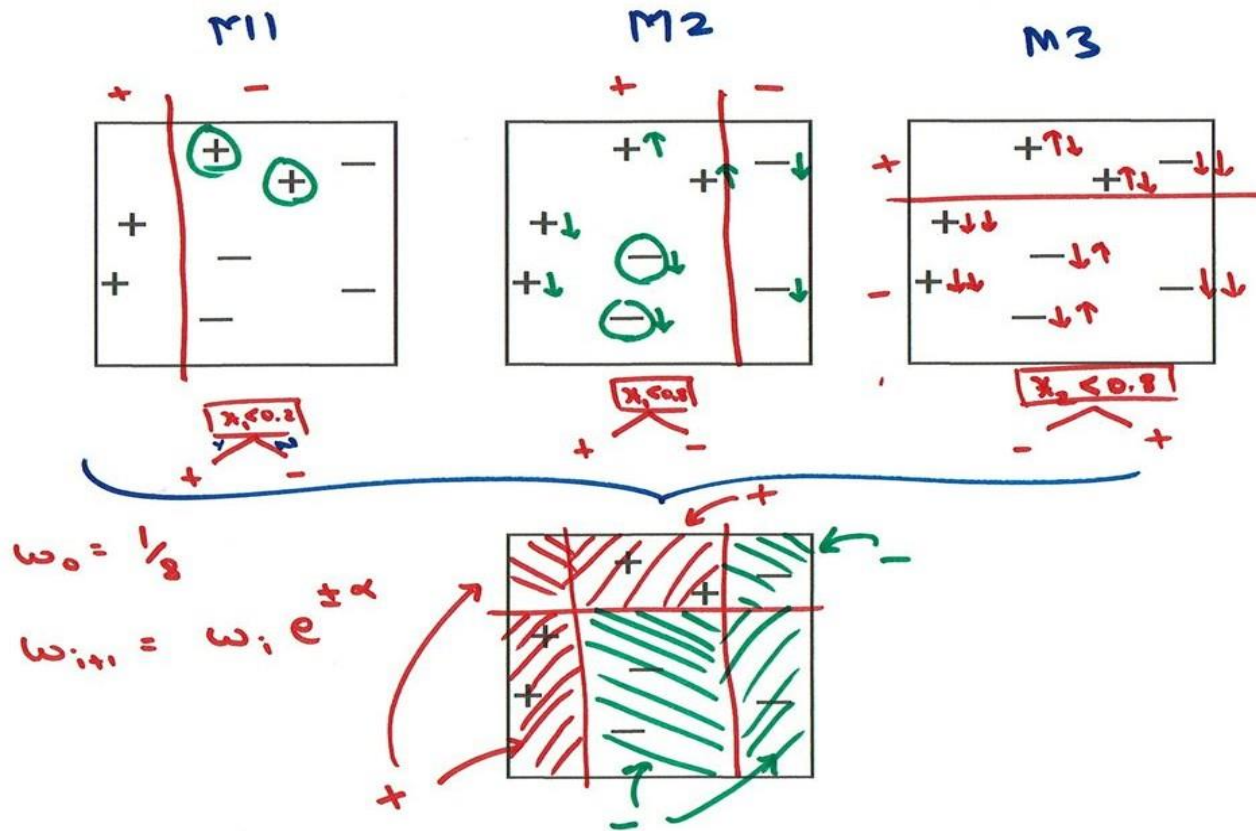
# Boosting Methods

- **AdaBoosting** (Adaptive Boosting)
  - In AdaBoost, the successive learners are created with a focus on the ill fitted data of the previous learner
  - Each successive learner focuses more and more on the harder to fit data i.e. their residuals in the previous tree
- **Gradient Boosting** (GBM)
  - Each learner is fit on a modified version of original data. Original data is replaced with the x values and residuals from previous learner
  - By fitting new models to the residuals, the overall learner gradually improves in areas where residuals are initially high
- **XG Boost (Extreme Gradient Boosting)** (XG M)
  - Upgraded implementation of Gradient Boosting. Developed for high computational speed, scalability, and better performance.
  - Parallel Implementation, Cross-Validation, Cache Optimization, Distributed Computation

# AdaBoost

| X1  | X2  | Y    |
|-----|-----|------|
| ... | ... | +    |
| ..  | ... | +    |
| ..  | ... | -    |
| ..  | ... | ...  |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |
| ..  | ... | .... |

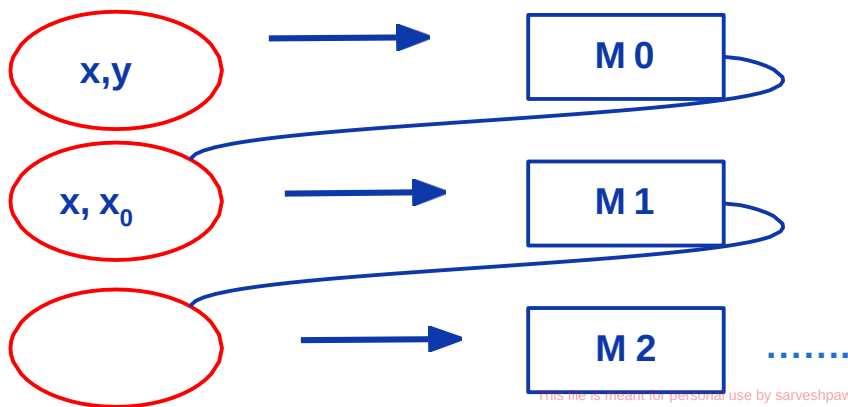
# AdaBoost



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# Gradient Boosting

|  | x |  | y     | $y_0$ | $y - y_0$ | h     |
|--|---|--|-------|-------|-----------|-------|
|  |   |  | 50    | 40    | 10        | 8     |
|  |   |  | 92    | 100   | -8        | -8    |
|  |   |  | 60    | 80    | -20       | -10   |
|  |   |  | 64    | 50    | 14        | 12    |
|  |   |  | ..... | ....  | .....     | ..... |



$$y_0 = h_0(x)$$

$$y = h_0(x) + \alpha_1 h_1(x)$$

$$y = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) + \dots$$

# Stacking

Weak  
Learners



Computationally  
Efficient



Bagging

Boosting

Stacking

Parallely, Independent

Homogeneous Model  
Combine using voting or avg

Seq, adaptively

Heterogeneous Model

Combine using another  
meta model

